

- RAG Learning

3.3 Decision-Making Techniques

Business — Edexcel (9BS0) — Theme 3: Business decisions and strategy

This is the most quantitative topic in Theme 3. Four techniques are named in the specification — **time-series sales forecasting**, **investment appraisal** (payback, ARR and NPV), **decision trees** and **critical path analysis** — and each is examined in two ways: a short calculation (typically 4 marks: Knowledge 1, Application 3) and an extended question asking whether the business should rely on the result. The second is where most marks are, so every technique below ends with its limitations. The recurring evaluative idea: **these techniques reduce uncertainty; they do not remove it**, and their outputs are only as reliable as the estimates fed into them.

Time-series analysis — forecasting by identifying patterns (trend, seasonal and cyclical variation) in past data recorded at regular intervals.

Moving average — an average of a fixed number of consecutive data points, recalculated as the period moves forward, used to smooth out fluctuations and reveal the trend.

Extrapolation — extending the trend of past data into the future to make a forecast.

Correlation — the strength of the relationship between two variables, shown on a scatter graph.

Investment appraisal — quantitative techniques used to assess whether a capital investment is financially worthwhile.

Payback period — the time taken for the net cash inflows from an investment to repay its initial cost.

Average (Accounting) Rate of Return (ARR) — average annual profit from an investment as a percentage of the initial investment.

Net Present Value (NPV) — the total discounted value of future net cash flows minus the initial cost of the investment.

Decision tree — a diagram showing the options, possible outcomes, probabilities and financial values of a decision, used to calculate expected values.

Critical Path Analysis (CPA) — a planning technique that identifies the sequence of activities that determines the minimum duration of a project.

1. Quantitative sales forecasting (3.3.1)

Why forecast?

Sales forecasts drive almost every other plan: production scheduling and capacity, stock ordering, workforce planning, cash-flow forecasts and budgets. An inaccurate forecast leads to overstocking (storage costs, waste, discounting) or understocking (lost sales, disappointed customers).

Moving averages — three-period

Add three consecutive values, divide by 3, and place the result against the **middle** period. Then move forward one period and repeat.

Month	Sales (£000)	3-period total	3-period moving average
Jan	62		
Feb	70	$62 + 70 + 66 = 198$	66.00
Mar	66	210	70.00
Apr	74	221	73.67
May	81	232	77.33
Jun	77	243	81.00
Jul	85	252	84.00
Aug	90	261	87.00
Sep	86		

The raw data zig-zags; the moving average rises steadily from 66.00 to 87.00 — an upward trend of about £3,500 a month $((87.00 - 66.00) \div 6 = 3.5)$.

Moving averages — four-quarter (centred)

With quarterly data, a four-quarter average removes seasonal variation because each average contains one of every season. But the average of four quarters falls *between* two quarters, so it must be **centred**: average each pair of consecutive four-quarter averages and place the result against the quarter in between.

Quarter	Sales (£000)	4-quarter total	4-quarter average	Centred average (trend)	Seasonal variation (actual – trend)
2023 Q1	420				
2023 Q2	310	1,520	380.00		
2023 Q3	280	1,560	390.00	385.00	–105.00
2023 Q4	510	1,590	397.50	393.75	+116.25
2024 Q1	460	1,610	402.50	400.00	+60.00
2024 Q2	340	1,660	415.00	408.75	–68.75
2024 Q3	300	1,700	425.00	420.00	–120.00
2024 Q4	560	1,730	432.50	428.75	+131.25

2025 Q1	500	1,760	440.00	436.25	+63.75
2025 Q2	370	1,800	450.00	445.00	−75.00
2025 Q3	330				
2025 Q4	600				

(The four-quarter totals sit between quarters: 1,520 is the total for 2023 Q1–Q4, placed between Q2 and Q3. Centred average for 2023 Q3 = $(380.00 + 390.00) \div 2 = 385.00$.)

Interpretation: the trend rises from 385.00 to 445.00 over seven quarters — about £8,570 a quarter ($60 \div 7 = 8.57$). Q4 is consistently the strongest quarter (Christmas) and Q3 the weakest.

Extrapolation and scatter graphs

Extrapolating the trend. To forecast 2026 Q1: the last trend value is 445.00 (2025 Q2); 2026 Q1 is three quarters later, so trend $\approx 445.00 + (3 \times 8.57) = 470.71$. The average Q1 seasonal variation is $(60.00 + 63.75) \div 2 = +61.88$. Forecast sales $\approx 470.71 + 61.88 = \text{£}532,600$ (to the nearest £100). Edexcel questions more often ask you to read a trend line off a graph and extend it, but the logic is identical.

Scatter graphs and the line of best fit. A scatter graph plots one variable against another (e.g. advertising spend against sales, or temperature against ice-cream sales). A line of best fit drawn through the points shows the relationship:

- **Positive correlation** — both rise together (advertising and sales).
- **Negative correlation** — one rises as the other falls (price and quantity demanded).
- **No correlation** — points scattered randomly.
- **Strength** — the closer the points lie to the line, the stronger the correlation.

Worked example: if a line of best fit through a firm's monthly data is Sales = £40,000 + 5 × advertising spend, then spending £12,000 on advertising predicts sales of $40,000 + (5 \times 12,000) =$

£100,000. Extrapolating to £50,000 of spend (predicting £290,000) would be unsafe if the data only covered spending of £5,000–£15,000 — the relationship may not hold (diminishing returns to advertising).

Exam Tip: Correlation does not prove causation. Sales and advertising may both rise because of a third factor (e.g. the summer season). Saying so is an easy evaluation point in any question built on a scatter graph.

Limitations of quantitative sales forecasting

- Assumes the future will resemble the past — breaks down with sudden external shocks (the Covid-19 lockdowns of 2020, the 2022 energy price spike, sudden tariff changes in 2025).
- Ignores qualitative factors: competitor actions, changes in tastes, new technology, social media trends.
- The further ahead the forecast, the less reliable it is.
- Past data may be unreliable or too limited; new products have no past data at all.
- Moving averages lag behind turning points in the data.
- Forecasts can be improved by combining with qualitative methods: market research, expert opinion, the Delphi technique, sales-force estimates.

2. Investment appraisal (3.3.2)

Worked data used below. A manufacturer is considering two machines, each costing £500,000 with a five-year life.

Year	Project A net cash flow (£000)	Project B net cash flow (£000)
0	(500)	(500)
1	150	100
2	180	120

3	200	150
4	170	200
5	100	230
Total inflows	800	800

Simple payback

Add up the net cash flows until the cumulative figure reaches the initial cost. For the part-year:

$$\text{Months} = (\text{amount still to be recovered} \div \text{net cash flow in that year}) \times 12$$

Year	A: cumulative cash flow	B: cumulative cash flow
0	(500)	(500)
1	(350)	(400)
2	(170)	(280)
3	+30	(130)
4	+200	+70

- **A:** after 2 years, £170,000 remains; year 3 inflow is £200,000. $170 \div 200 \times 12 = 10.2$ months.
Payback = **2 years 10.2 months** (2 years and about 10 months).
- **B:** after 3 years, £130,000 remains; year 4 inflow is £200,000. $130 \div 200 \times 12 = 7.8$ months.
Payback = **3 years 7.8 months** (3 years and about 8 months).

Strengths: simple, quick, understood by non-specialists; focuses on cash and liquidity; favours projects that reduce risk by recovering cash quickly — useful in fast-changing markets (technology) or for firms short of cash. **Weaknesses:** ignores all cash flows after payback (so ignores total profitability) and ignores the time value of money within the payback period.

Average (Accounting) Rate of Return

$$\text{ARR (\%)} = (\text{average annual profit} \div \text{initial investment}) \times 100$$

where $\text{average annual profit} = (\text{total net cash inflows} - \text{initial cost}) \div \text{number of years}$.

- **A:** total profit = $800 - 500 = \text{£}300,000$; average annual profit = $300 \div 5 = \text{£}60,000$; $\text{ARR} = 60 \div 500 \times 100 = \mathbf{12\%}$.
- **B:** identical totals, so $\text{ARR} = \mathbf{12\%}$.

The ARR can be compared with the firm's criterion rate (target return), the cost of borrowing, or the return on alternative uses of the money (e.g. interest on savings). If a loan costs 7% and ARR is 12%, the project covers its financing cost. **Strengths:** uses all cash flows over the project's life; gives a percentage comparable across projects of different sizes and with interest rates.

Weaknesses: ignores timing — here it rates A and B equally even though A returns cash much sooner; ignores the time value of money.

Discounted cash flow — Net Present Value

Money received in the future is worth less than money today, because today's money could be invested to earn interest (opportunity cost), and because of inflation and risk. Future cash flows are therefore multiplied by **discount factors** (always given in the exam) to convert them to **present values**.

$$\text{NPV} = \text{sum of (net cash flow} \times \text{discount factor)} - \text{initial cost}$$

Year	Discount factor (8%)	A: cash flow	A: present value	B: cash flow	B: present value
0	1.000	(500)	(500.00)	(500)	(500.00)
1	0.926	150	138.90	100	92.60
2	0.857	180	154.26	120	102.84
3	0.794	200	158.80	150	119.10
4	0.735	170	124.95	200	147.00

5	0.681	100	68.10	230	156.63
Total present value of inflows			645.01		618.17
NPV			+145.01 (£145,010)		+118.17 (£118,170)

Interpretation: a positive NPV means the investment earns more than the discount rate (the required return); a negative NPV means it should be rejected on financial grounds. Both projects are worthwhile, but **A has the higher NPV and the faster payback**. ARR could not separate them; NPV can, because A's cash arrives earlier and is discounted less.

Strengths: takes account of the time value of money and of all cash flows; gives a clear decision rule. **Weaknesses:** more complex to calculate and explain; the result depends heavily on the discount rate chosen — a higher rate lowers the NPV, so the choice can be manipulated; the NPV figure is not a percentage, so comparing projects of very different sizes is harder.

Summary of techniques

Technique	Measures	Time value of money?	All cash flows?	Best used when...
Payback	Speed of return / risk	No	No	Cash is tight; technology or markets change quickly
ARR	Profitability (%)	No	Yes	Comparing with a target return or interest rate
NPV	Value created in today's money	Yes	Yes	Long projects; interest rates are significant

Limitations of investment appraisal

- All three depend on **forecast cash flows**, which are estimates — and the further ahead, the less reliable. Managers keen on a project may be optimistic.

- They ignore **qualitative factors**: fit with corporate objectives, impact on staff morale, brand image, ethical and environmental effects, reactions of competitors.
- External change (inflation, interest rates, exchange rates, new regulation) can make the forecasts obsolete.
- The choice of discount rate (NPV) or criterion rate (ARR) is a judgement.
- Some investments are essential regardless of the numbers — e.g. meeting new safety law or replacing a failed IT system.

Exam Tip: Calculation questions are usually 4 marks (Knowledge 1 for the correct formula, Application 3 for the working). The 2024 mark schemes award full marks for a correct final answer alone, but give method marks if you go wrong — so always show working. Include units (£, %, years and months) and follow any instruction on decimal places: in 2024, an answer lacking the % sign or the required 2 decimal places lost a mark.

3. Decision trees (3.3.3)

Constructing a decision tree

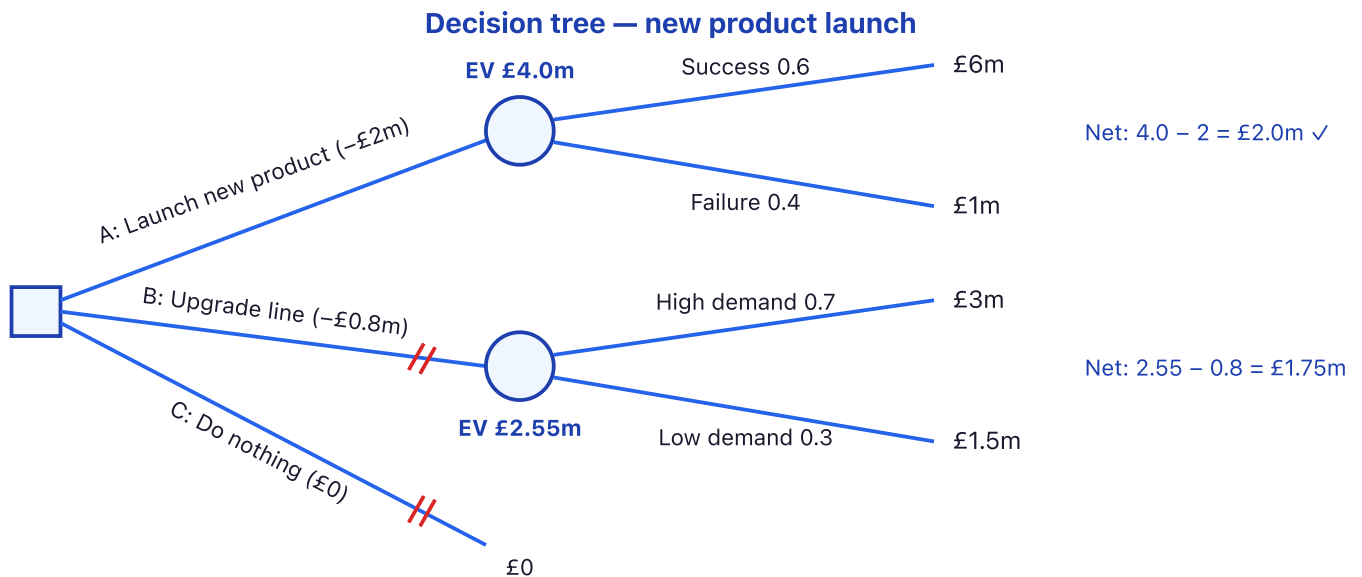
- **Square** = decision node — a point where the business chooses between options.
- **Circle** = chance node — a point where outcomes are uncertain; the probabilities on the branches from a chance node must add up to 1.
- **Branches** show options (with their costs) and outcomes (with probabilities and financial values).
- Work **right to left**: calculate the expected value (EV) at each chance node, subtract the cost of each option, and choose the option with the highest net EV. Rejected options are marked with a double line (//).

Expected value = sum of (probability × financial outcome)

Worked example

A food manufacturer can (A) launch a new product line at a cost of £2m, (B) upgrade an existing line at a cost of £0.8m, or (C) do nothing. Research suggests a new product has a 0.6 chance of

success (revenue £6m) and 0.4 chance of failure (£1m); the upgrade has a 0.7 chance of high demand (£3m) and 0.3 chance of low demand (£1.5m).



- **Option A:** $EV = (0.6 \times £6m) + (0.4 \times £1m) = £3.6m + £0.4m = £4.0m$. Net EV = $£4.0m - £2m = £2.0m$.
- **Option B:** $EV = (0.7 \times £3m) + (0.3 \times £1.5m) = £2.1m + £0.45m = £2.55m$. Net EV = $£2.55m - £0.8m = £1.75m$.
- **Option C:** £0.

Decision on financial grounds: launch the new product (A), with the highest net expected value of £2.0m.

Interpretation — the part that earns evaluation marks: the EV is an *average* that would be achieved if the decision were repeated many times; it will never actually occur. For A the real outcome is either +£4m ($6 - 2$) or -£1m ($1 - 2$). For B the worst case is still a gain of £0.7m ($1.5 - 0.8$). A **risk-averse** firm, or one with weak cash flow that could not survive a £1m loss, might rationally choose B despite its lower EV. The difference in EV (£0.25m) is small relative to the difference in risk.

Limitations of decision trees

- Probabilities and financial outcomes are **estimates** — often based on past experience or subjective judgement; a small change in probability can reverse the decision (here, if A's

success probability fell to 0.55, its net EV would be $(0.55 \times 6) + (0.45 \times 1) - 2 = \text{£}1.75\text{m}$, equal to B's).

- They reduce a decision to a few outcomes; reality has many.
- They ignore qualitative factors: fit with strategy, staff and brand impact, ethics.
- They ignore the time value of money unless the outcomes are discounted.
- They ignore the decision-maker's attitude to risk.

4. Critical Path Analysis (3.3.4)

Nature and purpose

CPA breaks a project into activities, identifies which depend on others and which can run **in parallel**, and calculates the **minimum project duration**. The **critical path** is the longest sequence of dependent activities through the network: any delay to a critical activity delays the whole project. Purposes:

- Identify the shortest possible completion time and plan a realistic deadline.
- Show which activities must be monitored closely (critical) and which have slack (float).
- Plan resources — order materials and schedule staff to arrive just when needed (links to lean production and just-in-time).
- Reallocate resources from non-critical to critical activities to shorten the project.
- Monitor progress: if a critical activity slips, managers know at once that action is needed.

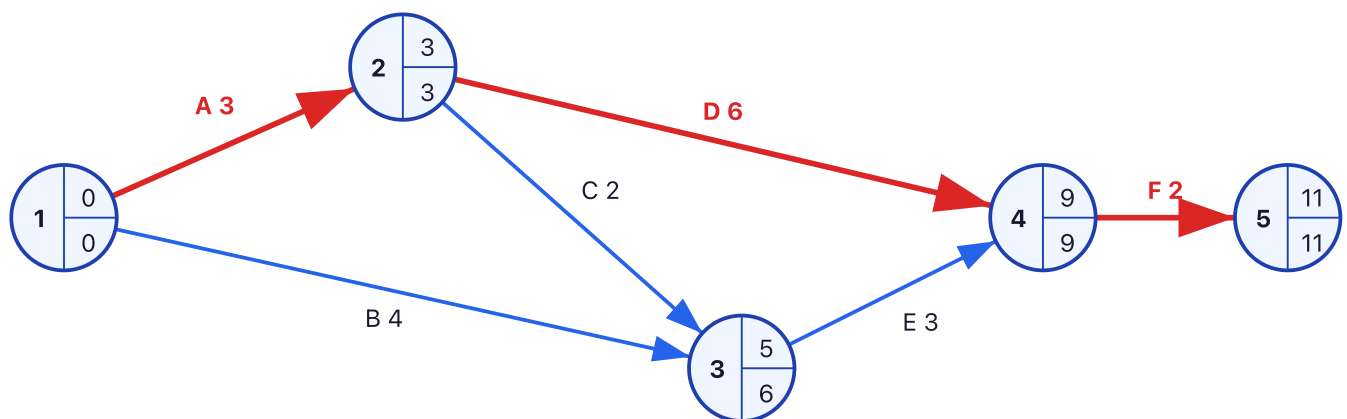
Drawing the network (Edexcel conventions)

- Activities are shown as **arrows** (lines) labelled with the activity letter and its duration.
- Events are shown as **nodes** (circles) divided into three: **node number** on the left, **Earliest Start Time (EST)** top right, and **Latest Finish Time (LFT)** bottom right.
- The network must start and end at a single node; arrows do not cross where avoidable.

Worked example — opening a new store

Activity	Description	Preceded by	Duration (weeks)
A	Design store layout	—	3
B	Order and deliver equipment	—	4
C	Recruit staff	A	2
D	Fit out premises	A	6
E	Train staff on equipment	B, C	3
F	Stock shelves and open	D, E	2

Network — critical path in red (A → D → F = 11 weeks)



Node key: node number (left) | EST (top right) | LFT (bottom right)

Calculating Earliest Start Times (forward pass)

Start at node 1 with EST = 0 and work left to right. EST of a node = EST of the previous node + duration of the activity leading into it. **Where two or more activities enter a node, take the highest value**, because the next activity cannot start until all preceding activities are finished.

- Node 2: $0 + A(3) = 3$
- Node 3: via B = $0 + 4 = 4$; via C = $3 + 2 = 5$ → take the higher: **5**
- Node 4: via D = $3 + 6 = 9$; via E = $5 + 3 = 8$ → **9**
- Node 5: $9 + F(2) = 11$ — the minimum project duration is 11 weeks.

Calculating Latest Finish Times (backward pass)

Start at the final node, where LFT = EST (11), and work right to left. LFT of a node = LFT of the following node – duration of the activity leaving it. **Where two or more activities leave a node, take the lowest value.**

- Node 4: $11 - F(2) = 9$
- Node 3: $9 - E(3) = 6$
- Node 2: via D = $9 - 6 = 3$; via C = $6 - 2 = 4 \rightarrow$ take the lower: **3**
- Node 1: via A = $3 - 3 = 0$; via B = $6 - 4 = 2 \rightarrow$ **0**

Total float

Total float = LFT (at end of activity) – duration – EST (at start of activity)

Activity	EST (start node)	Duration	LFT (end node)	Total float	Critical?
A	0	3	3	$3 - 3 - 0 = 0$	Yes
B	0	4	6	$6 - 4 - 0 = 2$	No
C	3	2	6	$6 - 2 - 3 = 1$	No
D	3	6	9	$9 - 6 - 3 = 0$	Yes
E	5	3	9	$9 - 3 - 5 = 1$	No
F	9	2	11	$11 - 2 - 9 = 0$	Yes

Critical path: A → D → F (3 + 6 + 2 = 11 weeks). Activities with zero float are critical; the critical path is also visible where EST = LFT at every node along it (nodes 1, 2, 4, 5).

Interpretation: equipment ordering (B) could be delayed by up to 2 weeks without delaying the opening, so its staff or budget could temporarily support the fit-out (D). If the fit-out overruns by one week, the store opens in week 12 — which could mean missing a key trading period and paying an extra week's rent and wages with no revenue.

Limitations of Critical Path Analysis

- Durations are **estimates**; over-optimistic estimates make the plan unreliable (large construction and IT projects routinely overrun — HS2 is a well-known UK example of cost and time overruns).
- It does not guarantee that the project will be completed on time — it only plans; managers still have to manage.
- Complex projects produce very complex networks that are hard to draw and update (software is used in practice).
- It assumes resources are available when needed and does not show resource constraints (e.g. the same team needed for two parallel activities).
- Focusing on time can encourage rushing, with quality or safety compromised.
- It gives no information on costs unless combined with a budget.

Exam Tip: Edexcel regularly asks you to complete a partly-drawn network (fill in missing ESTs and LFTs, or identify the critical path) or to calculate total float for one activity. The two rules to learn: forward pass — take the **highest** where arrows meet; backward pass — take the **lowest** where arrows split. Evaluation questions then ask whether CPA will ensure a project is completed on time: the answer is that it improves planning and control but depends on accurate estimates and good management.

How Edexcel rewards this topic

- **Knowledge:** correct formula (worth 1 mark on a 4-mark calculation) and correct terminology — “expected value”, “total float”, “time value of money”.
- **Application:** accurate calculation using the case figures; units and decimal places as instructed.
- **Analysis:** interpret the result — what does a 2-year 10-month payback or a positive NPV mean for this firm's cash flow and risk?
- **Evaluation:** balance the quantitative result against its reliability and against qualitative factors (objectives, stakeholders, ethics, market conditions); reach a judgement that states what the decision depends on.

Common mistakes

- Placing a moving average against the wrong period — it belongs to the **middle** period; four-quarter averages must be centred.
- Forgetting to subtract the cost of an option from its expected value in a decision tree.
- Probabilities from one chance node that do not add up to 1.
- Calculating ARR on total rather than *average annual* profit, or forgetting to deduct the initial cost first.
- Converting the payback fraction to months incorrectly (0.85 of a year is 10.2 months, not 8.5 months).
- Stating that a project with positive NPV “makes a profit of” that amount — NPV is the value added above the required return.
- Taking the lowest value on the forward pass (or highest on the backward pass) in CPA.

Quick recall

- 3-period MA: total of 3 ÷ 3, placed at the middle period; 4-quarter MA must be centred
- Extrapolation: extend the trend; add average seasonal variation; unreliable after shocks
- Scatter graphs: positive/negative/no correlation; correlation ≠ causation
- Payback: cumulative cash flow; part-year = (remaining ÷ year's inflow) × 12
- ARR = average annual profit ÷ initial investment × 100
- NPV = sum of discounted net cash flows – initial cost; positive = accept
- Decision tree: square = decision, circle = chance; EV = $\Sigma(\text{probability} \times \text{outcome})$; subtract cost
- CPA: forward pass (highest) gives EST; backward pass (lowest) gives LFT; total float = LFT – duration – EST; critical = zero float
- All techniques: estimates, ignore qualitative factors, reduce but do not remove uncertainty